

sparta.

SPectroscopic Absorption, Real Time Analysis

Concatenator · Absorbance Calculator · Plotter

Quick start guide

Getting data in, teaching SPARTA your file names,
and the recipes for the outputs you need.

v1.5.0

NSLS-II 22-IR-1 — Dr. Lee's Lab

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CONTENTS

Quick start	1
The naming system	4
Workflows	11
Keyboard shortcuts	15

The same text is in the program: left panel, Guide / notes, View dropdown. Hover any control for a tip; F1 lists the keyboard shortcuts.

16 pages

QUICK START

Quick start covers the path from raw segment files to a figure. SPARTA writes to disk on Run, Export data and Save only.

Point the Input folder at demo_data, which ships beside the program: 5 series points, 3 channels, 2 grating segments, 22-IR-1 naming, one point tagged _D. About > 'Welcome & tour...' loads that folder and walks this path in the window. Esc leaves the tour.

1. POINT AT THE DATA

Input folder, in the left panel: press Browse, drag a folder from Explorer onto the window, or paste a path and press Enter. Enter also rescans and remembers the folder.

The scan reads one folder: the folder holding the segment files themselves.

Output folder: press Browse and pick a writable folder. A Run creates <output>/<inputname>_absorbance/. If that subfolder exists and holds files, SPARTA appends a timestamp (<inputname>_absorbance_YYYYMMDD_HHMM).

2. CHECK HOW THE NAMES ARE READ

The button under the Input box shows the active profile. The default is '22-IR-1 default', which reads

```
vis_{DAC}_{Sample}[_{Pressure}][_bg|_s][_C|_D][_rep][.{seq}]
```

so vis_Y04_Arch29_26p0_s.003 is DAC Y04, sample Arch29, 26.0 GPa, the sample channel, grating segment 3.

For another scheme, open the button and press 'Guess format'. In the Preview, green means parsed and red means skipped with the reason. Correct the fields, then press 'Use this profile'. The Naming system view holds the walkthrough.

3. RUN

Press Run. SPARTA joins the grating segments of each measurement and computes

$$A = -\log_{10}[(\text{Sample} - \text{Dark}) / (\text{Background} - \text{Dark})]$$

A Run writes one CSV per measurement into the output subfolder, named {DAC}_{sample}_{value}[_C|_D]_absorbance.csv. Export tab > Data files ticks the columns that CSV carries; Defringed data is ticked by default. A _reduction.provenance.json sidecar carries the version, the parameters and every curve. The left panel's 'Export...' button, the Export tab's 'Export data...' and Ctrl+E all open the Export dialog, which writes the same columns into a folder you pick.

Each Progress log line is one measurement: OK lines carry the point count and the file written, SKIP lines the filename and the reason. Run reads Cancel during a run.

Absorbance takes all three channels. A measurement with fewer loads as raw counts, tagged in the trace name ([S only], [S+B]).

4. CHECK THE DATA

Every trace plots at once.

Inspect one trace in Plot tab > Plot mode. The left axis shows Sample, Background and Dark counts, the right axis the absorbance.

Traces, in the Data tab, holds one row per loaded point. A colored dot marks a quality check that fired; hover it for the reason (saturation, negative absorbance, missing channels).

Click a curve to select it, double-click to solo it, click a legend entry to hide it. Right-click one for inspect, solo, hide, toggle D, defringe compare and show in data table. Ctrl+D opens the data table, with 'Copy all (TSV)' and 'Open in Excel'.

5. MAKE THE FIGURE

Export tab > Figure: pick a Journal preset. One pick sets the publisher's column width and house style: typeface, text sizes, line weight, spines, ticks and DPI. 'Preview at export size (WYSIWYG)', in Export tab > Export, shows the printed proportions on screen.

Style tab > Legend: above about ten traces, turn on 'Direct labels at curves' or use Style tab > Colorbar.

Export tab > Export: 'Save plot...' writes PNG, PDF, SVG, EPS and TIFF. PDF and SVG are vector. 'Editable text' keeps the labels editable in Illustrator or Inkscape.

THE REST OF THE WINDOW

Session tabs above the plot hold independent sessions, each with its own data, folders, settings and undo history.

'Load previous run' reopens a finished output folder from its CSVs. Rescan (or F5) re-runs the folder after new files arrive.

The top bar carries Theme; the gear that opens the Settings panel (Font, Text size, Helper tips, Performance mode, Tutorial, About); and NUKE, which clears the session and asks first.

The Quick Access strip above the tabs starts empty; its gear picks what it carries.

F1 lists the keyboard shortcuts.

THE NAMING SYSTEM

A Run reads five things from every file: the DAC, the sample, the value of the experiment variable, the channel (sample / background / dark) and the grating segment.

TOKENS

A token is whatever sits between two separators.

```
vis_Y04_Arch29_26p0_s.003
 \ / \ / \ / \ / \ /
 1   2   3   4   5   segment suffix
```

Tokens 1 to 5 here are the prefix, the DAC id, the sample id, the value and the channel keyword. The segment suffix splits off before tokenizing, and empty tokens drop out.

A profile assigns a meaning to each token position, in order:

dac	the cell id. Required, unless a default gives it.
sample	the sample id. Required, unless a default gives it.
pressure	the value of the Series variable. Optional.
role	the channel keyword (sample / background / dark).
branch	the compression / decompression tag.
rep	the retake number.
ignore	a piece the pipeline does not need.

'pressure' is the internal field name; the dialog shows the Series variable's name.

THE TWO KINDS OF PROFILE

22-IR-1 default (built-in): a fixed grammar, read-only, stable across releases.

```
vis_{DAC}_{Sample}[_{Pressure}][_bg|_s][_C|_D][_rep][.{seq}]
```

- the name starts with 'vis' and carries at least three underscore-separated tokens
- a bare name = dark, _bg = background, _s = sample
- _C / _D is the branch tag, _2 / _3 a retake, in either order
- the value uses 'p' for the decimal: 1p39 = 1.39. A missing value token reads as 0, with a note in the log
- the segment suffix is a dot plus digits (.001 .. .00N). A file with no suffix is segment 1
- a trailing '.csv' twin is tolerated and deduplicated

```
vis_Y04_Arch29_26p0_s.003
  DAC      Y04
  sample   Arch29
  value    26.0
  channel  sample
  branch   none
  retake   1
  segment  3
```

Rejections:

```
vis_Y04_Arch29_26p0_s_r.003  unrecognized trailing token
vis_Y04_Arch29_26p0_sA.003   unrecognized trailing token
vis_Y04_Arch29_bg.001.002    malformed (extra extension)
vis_Y04_Arch29_26p0_s.abc    segment is not numeric
Y04_Arch29_26p0_s.003        does not match
                             vis_DAC_SAMPLE[_PRESSURE]
```

The built-in accepts bg / s / C / D / a digit retake. The fix is a custom profile or a per-file fix.

Custom profiles: everything above becomes editable, saved under a name and surviving restarts.

THE PROFILE FIELDS

Prefix

A fixed first token, matched case-insensitively and consumed ('vis' in the classic names). Another first token is skipped with "missing prefix '<x>'".

Separator

The text between tokens, usually _ or -. Comma-separate alternatives ('_','-'), tried longest-first.

Value decimal (labelled with the Series variable's name)

'p' reads 12p5 as 12.5, '.' reads 12.5, ',' reads 12,5. '.' and ',' are accepted whatever you pick.

Strip units

Unit text removed from the END of the value token, comma-separated and case-insensitive: with 'gpa,kbar', '15.3GPa' reads as 15.3.

Segment sep

The text before the segment number at the end of the name, one or more characters ('_seg' reads name_seg003). The RIGHTMOST occurrence splits base from segment.

Numbering

- digits: any all-digit run, read with `int()`, so .001, .01 and .1 all mean segment 1.
- letters: a = 1, b = 2 ... z = 26, then aa = 27 (spreadsheet style), case-insensitive. Capped at two letters, so .txt, .dat and .asc stay extensions.

No number =

What a bare name means: a segment index, normally 1, or 'reject', which skips the file with the reason "no segment suffix ('<sep><segment>' required)".

Background / Sample / Dark keyword(s)

The token that marks each channel, comma-separated alternatives (bg,ref), case-insensitive. Dark is the default role.

Compression / Decompression keyword(s)

The branch tag, shown as C and D on the plot, with D dashed. Left blank, both fall back to c/d.

Default DAC Name / Default Sample Name

For single-cell folders whose names omit that piece. Put the value here AND drop that label from the token order.

Token order

The meanings in the order the pieces appear, set by the chips under 'Teach by example'.

HOW A NAME IS PARSED, STEP BY STEP

1. A single trailing '.csv' is removed.
2. The segment suffix splits off: the text after the rightmost segment separator decodes under the numbering scheme. It counts as a segment only when it decodes AND something is left in front; otherwise the 'No number =' policy applies.
3. The rest tokenizes on the separator, and the prefix, if any, matches the first token and is consumed.
4. The token order walks left to right against the remaining tokens:
 - dac, sample: consume the token unconditionally. A missing token is an error ("missing dac token").
 - ignore: consume a token if one is there.
 - pressure: consume the token ONLY if it reads as a number, after unit-stripping and decimal substitution; otherwise it passes to the NEXT field.

- role: consume the token ONLY if it is a known channel keyword.
- branch: consume the token ONLY if it is a known branch keyword.
- rep: consume the token ONLY if it is all digits.

One token order therefore reads both vis_Y04_Arch29_bg.001 and vis_Y04_Arch29_26p0_bg_C_2.001.

- Any token left over is an error: "unrecognized trailing token".
- Defaults fill what the name omits: value 0, channel dark, retake 1, plus the DAC and sample defaults you set.

TEACH BY EXAMPLE

Open 'Name format' from the left panel.

- Pick a profile, or press 'Save as...' and name a new one. The built-in opens read-only.
- Set the grammar boxes: separator first, then prefix, decimal, unit stripping, keywords, segment convention.
- Under 'Teach by example', pick a filename that shows EVERY piece the scheme can produce. Each token gets a chip with a dropdown; set each dropdown to what that piece means. The gray pieces between chips are the literal separators, and the segment suffix sits at the end, captioned with how it was read. Click either to jump to the box that governs it.
- Watch the Preview, which parses every file in the folder with the grammar as it stands. Green rows parsed, red rows skipped with the reason, blue rows fixed by hand. The counter reads "matched N / M files". The preview reads at most 500 files.
- Press 'Use this profile'. A grammar with a problem makes the button refuse and lists it:
 - an empty separator
 - an unknown field
 - a role keyword that maps to no channel
 - dac or sample supplied by neither a token nor a default
 - an unusable segment scheme
 - a bad missing-segment value

GUESS FORMAT

'Guess format' reads the folder and proposes the whole grammar.
The preview is the real check.

It settles the segment convention first, scoring the candidate separators (. - __seg -seg) against both numbering schemes. The strong signal is one base recurring with SEVERAL segment values. All files numbered sets 'No number =' to 'reject', some files numbered sets 1.

It then takes the token separator with the most consistent token count, makes a shared literal first token the prefix, and classifies the rest as value, channel-keyword, branch and retake columns; the first two unclaimed columns become dac and sample. The keywords it knows are bg / ref / back / background, s / sam / samp / sample / sig, dark / dk / drk, c / comp / up, d / dec / decomp / down.

It reports "matched N / M files" as a toast and in the log. A folder with fewer than two id columns, or a profile the validator rejects, gets a plain default.

WORKED EXAMPLES

Every piece present, dash-separated:

```
vis-D42-fo90-15.3GPa-s-c-2.003
```

Prefix	vis
Separator	-
Value decimal	.
Strip units	gpa
Sample keyword	s
Compression	c
Segment sep	.
Numbering	digits
Order	dac, sample, pressure, role, branch, rep

DAC D42, sample fo90, 15.3, sample channel, compression, retake 2, segment 3.

Letter segments (a/b/c):

```
o1_run7_4p2_bg_b
```

Segment sep	_
Numbering	letters
No number =	1

The rightmost '_' splits 'b' off, which decodes as segment 2.
The segment separator and the token separator are the same character here; give segments their own separator where you can, such as '-b' or '_segb'.

Dash-numbered segments with underscore tokens:

```
quartz_C3_8p1_s-2

Separator      _
Segment sep    -
Numbering      digits
```

The conventions stay apart, because '-' serves the segment alone.

Single-cell folder, with DAC and sample outside the names:

```
12p5_bg.001

Prefix          (blank)
Separator       _
Default DAC Name Y04
Default Sample Name Arch29
Order           pressure, role
```

The output CSVs are still named
Y04_Arch29_12p5_absorbance.csv.

A spare piece: in vis_20260731_Y04_Arch29_26p0_s.003, label the date chip 'ignore'.

EDGE CASES AND TRAPS

Decimal character equal to the segment separator.
With decimal '.' and segment sep '.', a name ending in a dotted value ('..._2.5') has its '5' read as a segment, because the segment split runs first. Give segments a different separator, or use 'p'.

Token separator equal to the segment separator.
Legal. Any trailing token that decodes under the numbering scheme becomes the segment: under 'digits' a trailing retake number, under 'letters' a trailing one-letter or two-letter keyword.

A value of 0.
Allowed and meaningful. A MISSING value token also reads as 0, and the log then records "no pressure in filename, assumed 0 GPa".

Negative or non-finite values.

SPARTA takes a finite value at or above 0.

Case.

Prefix, channel keywords, branch keywords, unit suffixes and letter segments match case-insensitively. DAC and sample ids keep the case as written, because they become file names.

Two profiles, one folder.

One profile is active at a time. Run the majority convention, and fix the minority by hand.

FIXING STUBBORN FILES

Double-click a preview row, or press 'Fix selected...', to type its fields by hand: Channel role (dark / background / sample), DAC, Sample, <Series variable>, Replicate and Segment.

A blank field keeps what the parser produced. A fix beats any pattern, and a fix that supplies a channel role resurrects a rejected file.

'Exclude selected' leaves the file out of the run, and the note reads "excluded by user". 'Remove fix' undoes one file, 'Clear all fixes' the whole folder. Fixes are stored per folder and applied after parsing on every Run, Rescan and preview.

WHERE THE PARSED VALUES GO

dac, sample and the value become the output file name stem:
{DAC}_{SAMPLE}_{VALUE}[C_D]_absorbance.csv.

The trace's identity label is "{DAC} {SAMPLE} {VALUE} GPa", plus a branch tag and a retake note. The display relabels it to the active Series variable's unit; the label itself holds, as the key that presets, sessions and exports resolve through.

WORKFLOWS

Recipes by goal, each assuming a folder is loaded.

A PUBLICATION OVERLAY FIGURE

1. Run the folder and check the trace count in the log.
2. DATA > TRACES: tick the traces for the figure. 'Lock colors to all datasets' in STYLE > COLORS & COLORMAP holds the colors.
3. EXPORT > FIGURE: pick the journal preset for your column width.
4. EXPORT > EXPORT: 'Preview at export size (WYSIWYG)' shows the figure at its printed size.
5. Up to about ten traces use STYLE > LEGEND, Location 'outside right'; above that, STYLE > COLORBAR or 'Direct labels at curves'.
6. STYLE > TITLE & AXIS LABELS: override the default labels for a house form. Mathtext works: λ (nm), Fe^{2+} .
7. Save the plot as PDF with 'Editable text' on; 'Also save' adds PNG, and 'Grayscale copy' a mono version.
8. 'Export settings', under Progress, prints the configuration into the log for the methods section.

DEFRINGED COLUMNS FOR IGOR OR ANOTHER TOOL

1. Tick df beside the plot to see the cleaned curves; the df switch changes the plot only.
2. Read the fringe report in the log: the traces with a detected fringe, the fitted $n \cdot t$ in micron, and the p-value.
3. Check one trace with 'Defringe compare (selected trace)' in PLOT > 2D PLOT OPTIONS, then click a curve; the gray dashed line behind it is the pre-defringe absorbance. Narrow the half-width in FRINGE > FFT REMOVAL > Notch list when the notch took out real structure, widen it when fringes survive.
4. EXPORT > DATA FILES: Defringed data is ticked by default, so a Run writes the columns. For a folder already loaded, press 'Export...' in the left panel, or Ctrl+E, and Export in the dialog.
5. Each trace's CSV then carries Absorbance_notch, Background_notch and Sample_notch after the base columns. Each trace follows its own notch list, the rest the global controls.

SMOOTHED COLUMNS

Tick Smoothed data in EXPORT > DATA FILES, then Run or open the Export dialog. Each trace's CSV then carries Absorbance_smoothed, and Absorbance_notch_smoothed while Defringed data is on. Tune the filter first in DATA > SMOOTHING > 'Smoothing settings...'. Crop, in the Export dialog, limits one export to a wavelength range.

THICKNESS VERSUS PRESSURE

1. Tick df beside the plot. The detection supplies $n \cdot t$.
2. Check the fringe report; the 'mark misses' triangles show the traces without a detection.
3. PLOT > PLOT MODE: switch to Thickness (fringe $n \cdot t$). Sample points are filled circles, background points open squares; the S / B boxes pick the channels.
4. For solved physical thickness, take the traces through the fringe workbench: describe the stack, assign roles, Solve, Record point. 'Results vs pressure...' in FRINGE > SESSION plots the solved values.

COMPARING COMPRESSION AND DECOMPRESSION

1. Set the branches. An explicit _C / _D tag in the filename wins, and ten historical experiments are recognised. Otherwise tick the D box per trace in DATA > TRACES.
2. 'Decompression list...' in DATA > TRACES reads a CSV or text file of decompression values: a pressure_GPa column or plain numbers, 'p' allowed as the decimal. Each value takes the nearest loaded trace within 0.05.
3. Read the branches apart with 'Only C' and 'Only D'.
4. Overlay both. The Decompression traces controls in PLOT > 2D PLOT OPTIONS draw the D branch dashed under 'Style D traces apart', with its own width, opacity and markers.
5. STYLE > LEGEND: rename 'C' and 'D' in the two small boxes. The rename is display only; the file names and the D list keep C and D.
6. 'Export D list (CSV) by selection' writes the branch assignment back out.
7. Tick C/D tag in file name in EXPORT > DATA FILES to put the branch letter into every output file name. There is no second copy and no subfolder.

A CUSTOM QUANTITY, PLOTTED AND EXPORTED

1. DATA > FORMULAS: press New..., name it and give it a unit. The name labels the Y axis and the CSV column.
2. Write the expression over the loaded columns. A problems list and a live min / max / NaN preview update as you type; Save stays disabled until the formula is clean.
3. Tick df when the formula uses Sf, Bf or Af, and 'Show smoothed' when it uses As. A missing column is reported and leaves a gap.
4. Save, then click the dot beside the row. The formula becomes the Y axis ('formula: <name>').
5. 'Save formula CSVs...' writes one two-column file per trace: wavelength plus the formula, with the expression in the header comments. Ticking Formula values in EXPORT > DATA FILES adds the formula as one column of each trace's CSV instead, on every Run and every export.

LIVE ACQUISITION AT THE BEAMLINE

1. Run once as soon as there are a few measurements.
2. PLOT > PLOT MODE: turn on Auto rescan, at roughly your measurement cadence.
3. The status line under the plot reads 'auto-rescan: N s' while the timer is armed; a tick that finds new files re-reduces the folder.

COMPARING TWO DATASETS SIDE BY SIDE

1. Press Ctrl+T for a new session tab, with its own data, folders, settings and undo history.
2. Load or Run the second dataset there. The tab names itself after the folder, and a double-click renames it.
3. Ctrl+Tab cycles. Style one tab, save it as a preset in EXPORT > PRESETS & PROJECTS, and load it in the other.

COMING BACK TO A SESSION LATER

The curves alone: 'Load previous run...' on the left panel, pointed at the output subfolder a Run wrote.

The styling: EXPORT > PRESETS & PROJECTS, pick the preset and Load. A preset carries the plot controls only.

Everything: 'Open project...', which is every setting plus the input and output folders, in one .json.

KEYBOARD SHORTCUTS

Files and figures

Ctrl+Enter	Run
Ctrl+S	Save plot
Ctrl+E	Open the Export data dialog
Ctrl+Shift+C	Copy figure to clipboard
F5	Rescan the input folder
Enter (input)	Rescan, and remember that folder

Editing

Ctrl+Z	Undo
Ctrl+Y	Redo (also Ctrl+Shift+Z)

Session tabs

Ctrl+T	New tab (blank session)
Ctrl+W	Close the current tab
Ctrl+Tab	Next tab (Ctrl+Shift+Tab = previous)

Plot, with the mouse

Click curve	Select it
Dbl-click	Solo that curve (again = restore)
Click legend	Hide / show that trace
R-click curve	Quick actions
Drag a box	Zoom into it (2D)
Mouse wheel	Zoom (2D about the cursor; 3D camera)
Dbl-click pill	Reset that pane divider

The quick actions are inspect, solo, hide, toggle decompression (D), defringe compare, show in data table and deselect.

Plot, with the keyboard

Arrow keys	Pan the 2D view / orbit the 3D camera
+ / -	Zoom (2D about center; 3D camera)
0	Fit the 2D view / reset the 3D camera
1 / 2 / 3 / 4	Stacked & 3D: off / 2D stacked / 3D ridge / 3D shape
[/]	Previous / next colormap
Ctrl+R	Reset 3D view

Data

Ctrl+D	Toggle the raw data table
Ctrl+C (table)	Copy the selected rows

Fringe workbench

PgUp / PgDn	Previous / next pressure point
F11	Fill the screen in the pop-out (Esc leaves it)

Elsewhere

Drop a folder	Make it the input folder and rescan (Windows)
F1	This shortcuts list

Esc Close the dialog on top

Esc (running) Cancel the run

Single keys (numbers, brackets, arrows, + / - and 0) do nothing while you type in a box. Esc also cancels a pane-divider drag.